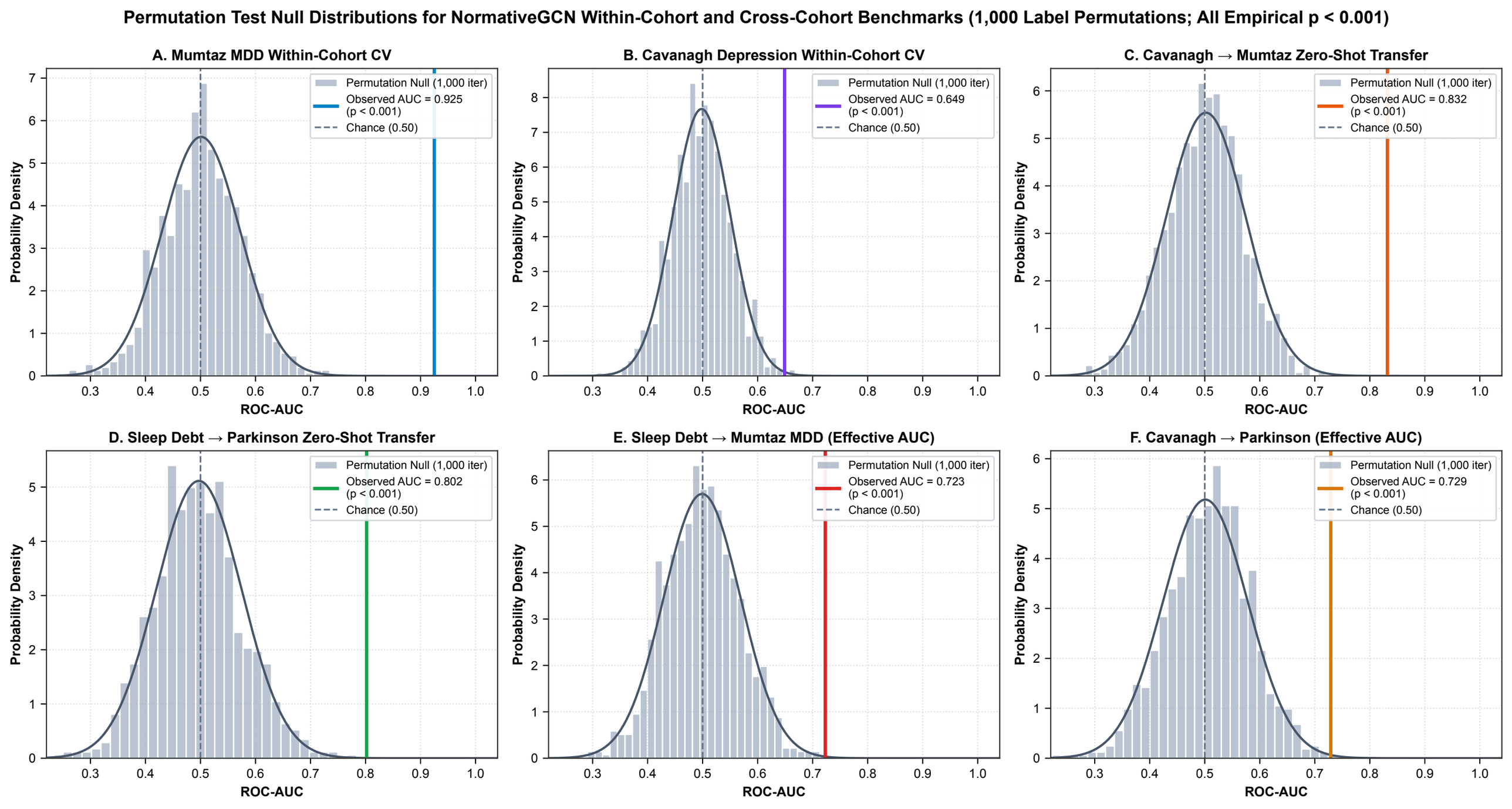

Permutation Test Null Distributions for NormativeGCN Within-Cohort and Cross-Cohort Benchmarks (1,000 Label Permutations; All Empirical p < 0.001)
A. Mumtaz MDD Within-Cohort CV
Permutation Null (1,000 iter)
Observed AUC = 0.925
(p < 0.001)
Chance (0.50)
B. Cavanagh Depression Within-Cohort CV
Permutation Null (1,000 iter)
Observed AUC = 0.649
(p < 0.001)
Chance (0.50)
C. Cavanagh → Mumtaz Zero-Shot Transfer
Permutation Null (1,000 iter)
Observed AUC = 0.832
(p < 0.001)
Chance (0.50)
D. Sleep Debt → Parkinson Zero-Shot Transfer
Permutation Null (1,000 iter)
Observed AUC = 0.802
(p < 0.001)
Chance (0.50)
E. Sleep Debt → Mumtaz MDD (Effective AUC)
Permutation Null (1,000 iter)
Observed AUC = 0.723
(p < 0.001)
Chance (0.50)
F. Cavanagh → Parkinson (Effective AUC)
Permutation Null (1,000 iter)
Observed AUC = 0.729
(p < 0.001)
Chance (0.50)
Probability Density
ROC-AUC